CONFIDENTIAL

FarmSense

The Deterministic Farming Operating System

29+	20-30%	$500/AF
PATENTS FILED	WATER SAVINGS	PUMPING FEE DEFENSE

The $500/AF Problem Meets Legal-Grade Precision

The Crisis

- → Colorado San Luis Valley: 89,000 AF/year aquifer depletion
- → $500/acre-foot groundwater pumping fee = 4x previous costs
- → Farmers face existential threat without data-driven defense
- → Water Court requires empirical proof, not estimates

The Solution

- → Deterministic OS: Rule-based, explainable logic (not AI black-box)
- → Digital Water Ledger: Court-admissible, cryptographically signed
- → 50m→1m resolution tiers: Free baseline to Enterprise legal fortress
- → 20-30% water reduction = $25,200/year savings per 126-acre pivot

The Opportunity

- → Subdistrict 1: 1,280 pivots, $7.6M ARR at scale
- → Subsidized hardware model: Free 50m tier ensures 100% participation
- → DoD dual-use: Environmental security, tactical trafficability intel
- → Global expansion: 5% irrigated cropland = 35M acres target

Subsidized Infrastructure + SaaS Upsell

The "Compliance Hook" Strategy

Governments and philanthropic partners subsidize hardware deployment to ensure 100% market participation. FarmSense monetizes through tiered SaaS, with Enterprise tier cross-subsidizing free baseline infrastructure.

$0
FREE TIER — 50M RESOLUTION

$49/mo
BASIC — 20M RESOLUTION

$199/mo
PRO — 10M + VRI ACTUATION

Custom
ENTERPRISE — 1M LEGAL PROOF

$50.80
TARGET LRZ COST AT SCALE

$117M
EST. 5% GLOBAL DEPLOYMENT

FUNDING SOURCE	AMOUNT	DEPLOYMENT
Gates Foundation (COP30)	$50M	Smallholder climate adaptation
DoD ESTCP	$25M	Base resilience, dual-use validation
World Bank G2G	$25M	Trans-boundary water treaties
Enterprise SaaS	$17M	Cross-subsidy from commercial

Proven Results & Regulatory Momentum

Validation Metrics

Kriging Accuracy	<5% Error
Mesh Uptime	99.5%
Water Savings	20-30%
ROI Increase	18-22%

Regulatory Milestones

- → June 2026: Water Court evidence submission (CSU SLV Pilot)
- → Q3 2026: DWR "Presumed Compliant" workflow integration
- → 2027: Colorado River Compact standardization
- → 2028: USDA/USGS Federal Water Credits partnership
- → 2029+: International G2G treaty compliance

Grant Pipeline

- → Federal ESG / DoD ESTCP: March 26, 2026
- → NSF SBIR Phase I: $275K (Q2 2026)
- → CWCB Water Plan Grant: $500K-$2M
- → BoR WaterSMART: $300K

Deterministic OS + Cryptographic Water Ledger

Sensor Hierarchy

- **LRZ (Lateral Root-Zone Scout):** $67.80 OEM, 1 per 15 acres
- **VFA (Vertical Field Anchor):** 48" deep-profile truth node
- **PMT (Pivot Motion Tracker):** Edge-compute hub, 50m grid
- **PFA (Pressure & Flow Anchor):** Wellhead safety actuator
- **DHU (District Hub):** 10km radius mesh manager
- **RSS (Regional Superstation):** 40ft container, 1m Kriging

Security & Privacy

- 128-bit AES encryption for field data
- FHSS (Frequency-Hopping) for LPI/LPD communications
- Dual-layer spatial privacy (exact GPS locked)
- 30-Day "Black Box" cache for offline resilience
- PBFT AllianceChain blockchain for audit trail
- FHE (Fully Homomorphic Encryption) planned

Dual-Use Capabilities

- JADC2 adapters for inter-agency data fabric
- Tactical trafficability scoring from soil moisture
- Trans-boundary water treaty enforcement
- Environmental security monitoring
- Low-bandwidth satellite backhaul for denied areas

TAM/SAM/SOM & Global Expansion

$12B
TAM: GLOBAL PRECISION AG

$2.4B
SAM: WATER-STRESSED REGIONS

$380M
SOM: COLORADO RIVER BASIN

5% Global Irrigated Cropland Target

700M acres global irrigated cropland (FAO) → 35M acres (5%) target. At 1 sensor per 15 acres = 2.3M units. $117M deployment cost fully funded via philanthropic + defense + commercial cross-subsidy. No dilutive equity required for global scale.

Phase 1-2: Prove (2026)

- → CSU SLV 2-Field Pilot
- → Water Court evidence <5% error
- → DoD ESTCP grant validation
- → $12,450 pilot hardware cost

Phase 3-5: Scale (2027-2035)

- → 100% Subdistrict 1 (1,280 pivots)
- → Colorado statewide → National → Global
- → Gates Foundation smallholder advisory mesh
- → International G2G treaty compliance

Capturing 5% of World Irrigated Cropland

300M

HECTARES GLOBAL IRRIGATED CROPLAND

15M

HECTARES (5%) — FARMSENSE TARGET

The Subsidized Penetration Model

- → Governments provide **free 50m tier** for compliance
- → Enterprise tiers ($30/acre) subsidize smallholder hardware
- → **$50.80 LRZ target** for emerging markets
- → 100% market participation via subsidy structure

Revenue at Scale

- → 15M hectares × $30/acre = **$11.1B TAM**
- → Conservative 10% penetration = $1.1B ARR
- → Hardware sales: $47B cumulative CapEx
- → Water trading fees: Additional 15% revenue

The Math: 5% of global irrigated cropland is achievable within 10 years through the subsidized "Compliance Hook" model — governments fund the free tier, enterprise farmers fund global expansion.

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03 / TRACTION

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Leadership

Jeremy
CEO / Systems Architect

Johnathon
CTO / Edge Compute

Fabian
Chief Scientist / Kriging

Advisory & Partners

CSU San Luis Valley Research Center (pilot site), Rio Grande Water Conservation District (regulatory partner), Colorado Division of Water Resources (compliance pathway), Department of Defense ESTCP (funding + dual-use validation).

Seed Round & Use of Funds

Seed Round: $1.01M

CATEGORY	ALLOCATION
Pilot Hardware (2 fields)	$60K
Engineering (6 months)	$400K
Grant Writing & Compliance	$100K
Regulatory & Legal	$150K
Operations & Contingency	$300K

18-Month Roadmap

- → **Q1 2026:** Federal ESG grant submission (March 26)
- → **Q2 2026:** CSU SLV pilot deployment
- → **June 2026:** Water Court evidence validation
- → **Q3 2026:** DoD ESTCP award ($2-5M)
- → **Q4 2026:** Gates Foundation COP30 framework
- → **Q1 2027:** Series A with strategic co-investors

Non-Dilutive Pipeline

- → Federal ESG / ESTCP: $2-5M
- → NSF SBIR Phase I/II: $275K → $1M
- → CWCB Water Plan: $500K-$2M
- → BoR WaterSMART: $300K
- → Gates Foundation: $50M target
- → Total Non-Dilutive: $55M+

Build the Global Water Ledger

Subsidized infrastructure. Court-admissible data. Sovereign water security.
Join the $1.01M seed round to validate the deterministic future of agriculture.

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LOCATION
San Luis Valley, Colorado

TRIAL DATE
June 29, 2026